

NOT STOCK CAN 1.85: power and CAN board behind the Waveshare ESP32-S3-Touch-LCD-1.85

Input: fuse, reverse polarity, load dump

VBAT 1
GND 2
CANH 3
CANL 4

J1 OBD 12V GND H L

VBAT_F VBAT

F1 0.5A 30V

VIN D1 SMD F

D2 SMD 26A

C1 VIN 4.7u 50V

C2 VIN 100n 50V

CAN transceiver, ESD, termination (JP1 open: the car's bus is terminated already)

TXD 1
RXD_T 4
+3V3 5
GND 8

U2 CANH TJA1051T/3

GND 2

+5V 3

C6 +5V 100n

C7 +3V3 100n

JP1 CANH CANL

R4 62R

R5 62R

TERM

12 V -> 5 V

VIN 5
EN 4
GND 2

U1 LMR16000

CB 1
SW 2
FB 3

R1 EN 100k

C3 CB 100n

D3 SW PNP 10CEH

L1 +5V 22u

R2 FB 56k

To the display ESP32-S3-Touch-LCD-1.85: UART socket (SH 1.0 cable, 1:1) and USB-C (5 V)

TXD 1
RXD 2
+3V3 3
GND 4

J2 to display UART

RX_D TXD

R6 1k

CC1 +5V

CC2 +5V

R7 56k

R8 56k

Mounting: the display's M2 holes

H1 M2

H2 M2

H3 M2

Supply flags

GND

VIN

+5V

+3V3

VBAT

12 V -> 5 V, CAN (TJA1051T/3) for the round gauge

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GND 2
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VBAT_F VBAT

F1 0.5A 30V

VIN D1 SMD F

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C1 4.7u 50V

C2 100n 50V

VIN

GND

VIN

GND

+5V

R1 100k

EN

VIN

GND

CB

C3 100n

SW

D3 SMD 10CEH

L1 22u

+5V

R2 56k

FB

GND

TXD 1
RXD 2
+3V3 3
GND 4

J2 to display UART

RX T

R6 1k

CC1

CC2

A5 CC1
B5 CC2

A7 CC1
B7 CC2

A6 CC1
B6 CC2

SBU1 A8
SBU2 B8

GND S1
GND A1

SHIELD

USB

5V

R7 56k

CC1

+5V

R8 56k

CC2

+5V

TMD

C8 4.7n

TERM

JP1

T/CANL

U2 CANH
TJA1051T/3
CANL

7 CANH
6 CANL

3 +5V
2 GND

1 TXD
4 RXD
5 +3V3
8 GND

12 V -> 5 V

VIN 5
EN 4
GND 2

U1 LMR16000

CB 1
SW 2
FB 3

CR

12 V -> 5 V, CAN (TJA1051T/3) for the round gauge

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H1 M2
H2 M2
H3 M2

Supply flags

GND

VIN

+5V

+3V3

VBAT

Mounting: the display's M2 holes

CAN transceiver, ESD, termination (JP1 open: the car's bus is terminated already)

12 V -> 5 V, CAN (TJA1051T/3) for the round gauge

NOT STOCK CAN 1.85: power and CAN board behind the Waveshare ESP32-S3-Touch-LCD-1.85

Input: fuse, reverse polarity, load dump

12 V -> 5 V

Supply flags

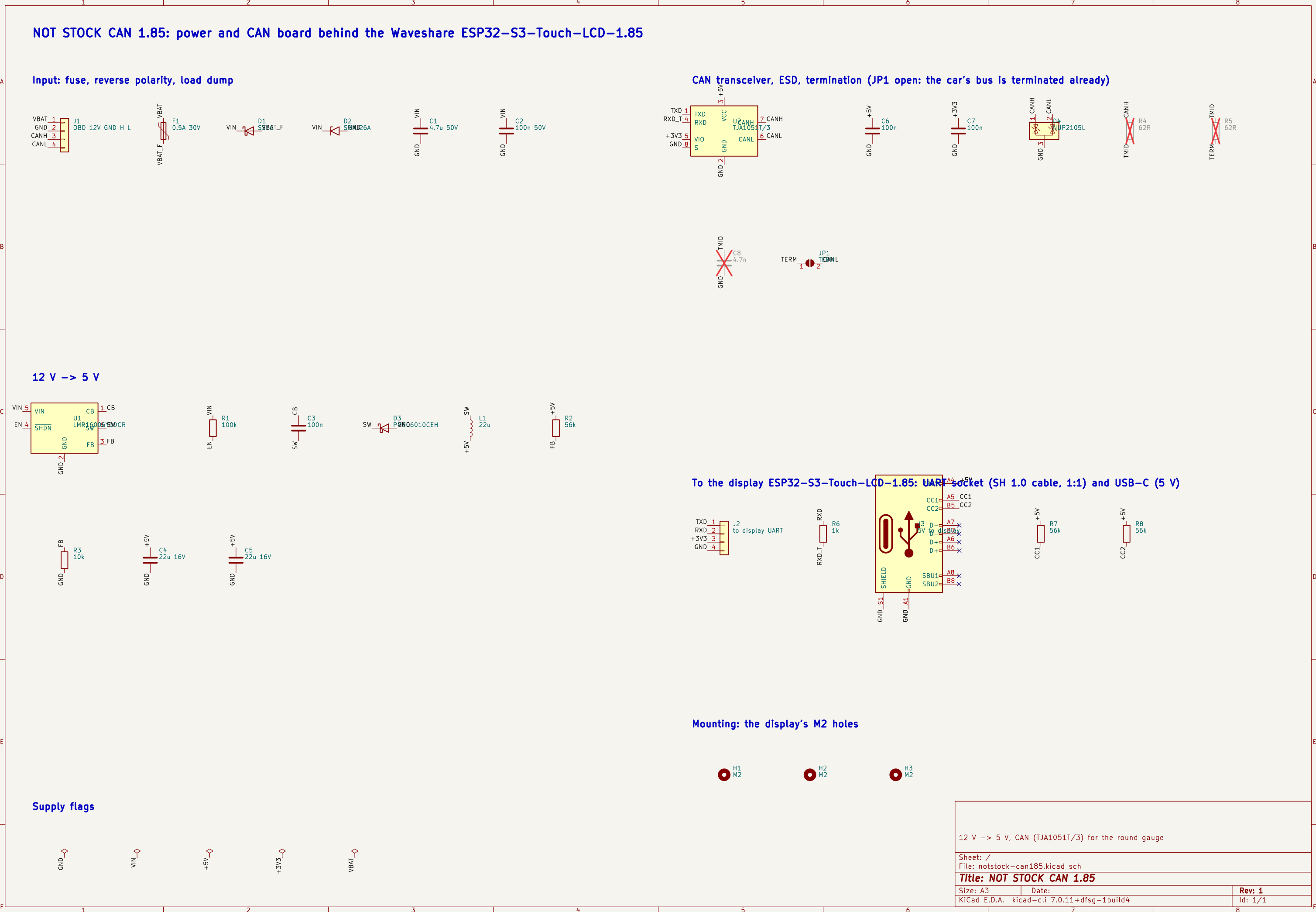
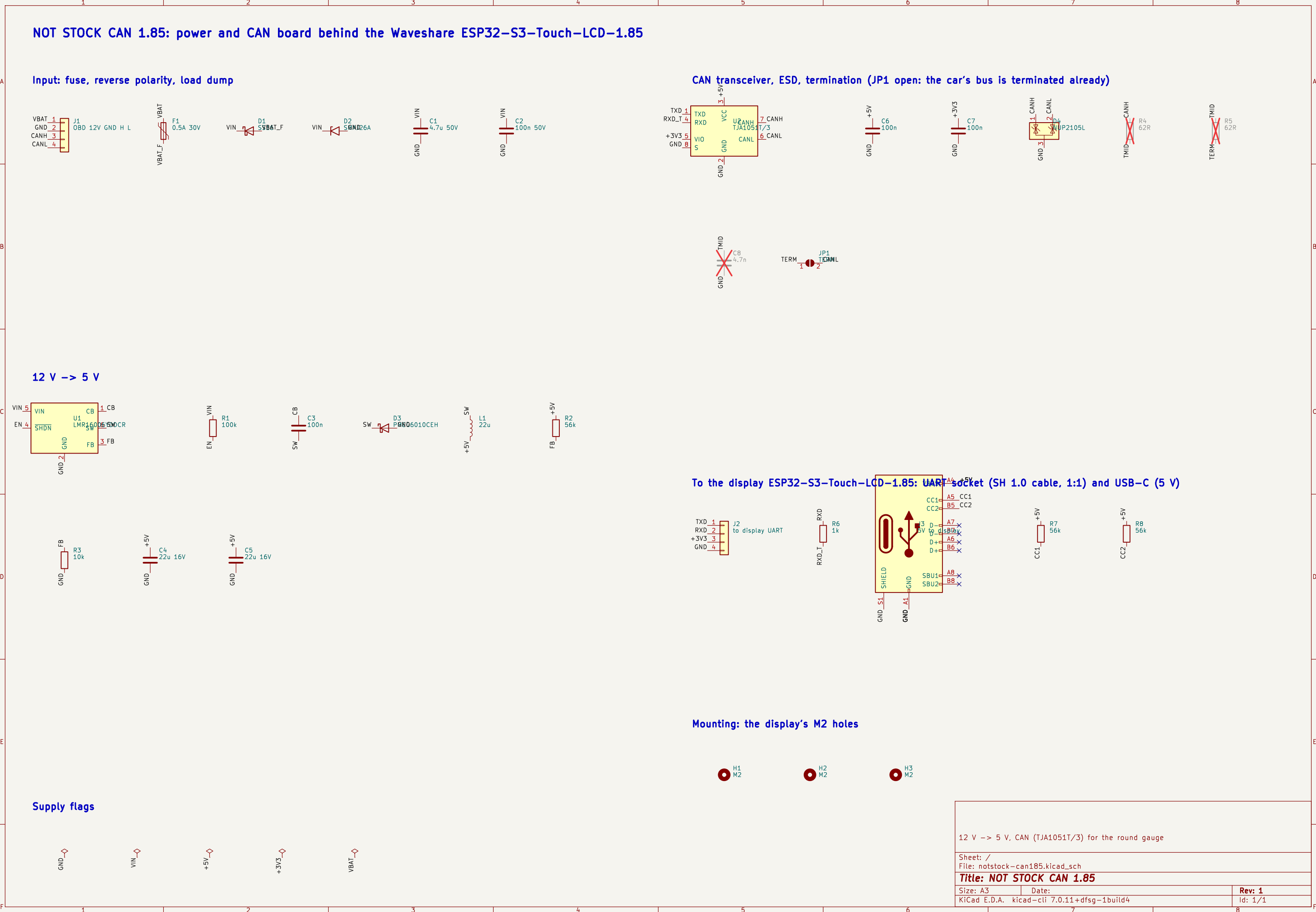
Mounting: the display's M2 holes

CAN transceiver, ESD, termination (JP1 open: the car's bus is terminated already)

To the display ESP32-S3-Touch-LCD-1.85: UART socket (SH 1.0 cable, 1:1) and USB-C (5 V)

12 V -> 5 V, CAN (TJA1051T/3) for the round gauge

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NOT STOCK CAN 1.85: power and CAN board behind the Waveshare ESP32-S3-Touch-LCD-1.85

Input: fuse, reverse polarity, load dump

CAN transceiver, ESD, termination (JP1 open: the car's bus is terminated already)

12 V -> 5 V

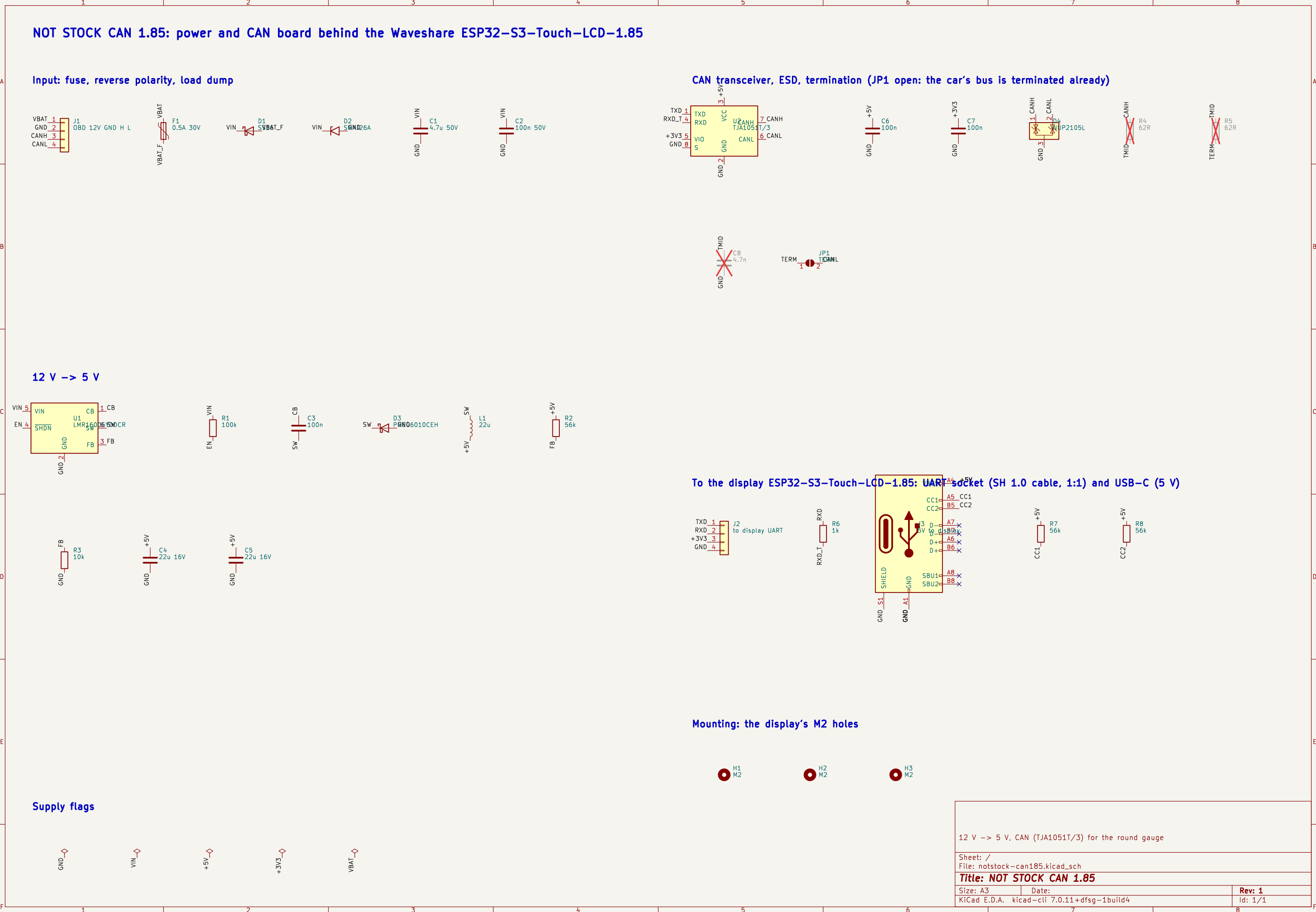
To the display ESP32-S3-Touch-LCD-1.85: UART socket (SH 1.0 cable, 1:1) and USB-C (5 V)

Mounting: the display's M2 holes

Supply flags

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NOT STOCK CAN 1.85: power and CAN board behind the **Waveshare ESP32-S3-Touch-LCD-1.85**

Input: fuse, reverse polarity, load dump

CAN transceiver, ESD, termination (JP1 open: the car's bus is terminated already)

12 V -> 5 V

To the display **ESP32-S3-Touch-LCD-1.85**: UART socket (SH 1.0 cable, 1:1) and USB-C (5 V)

Mounting: the display's M2 holes

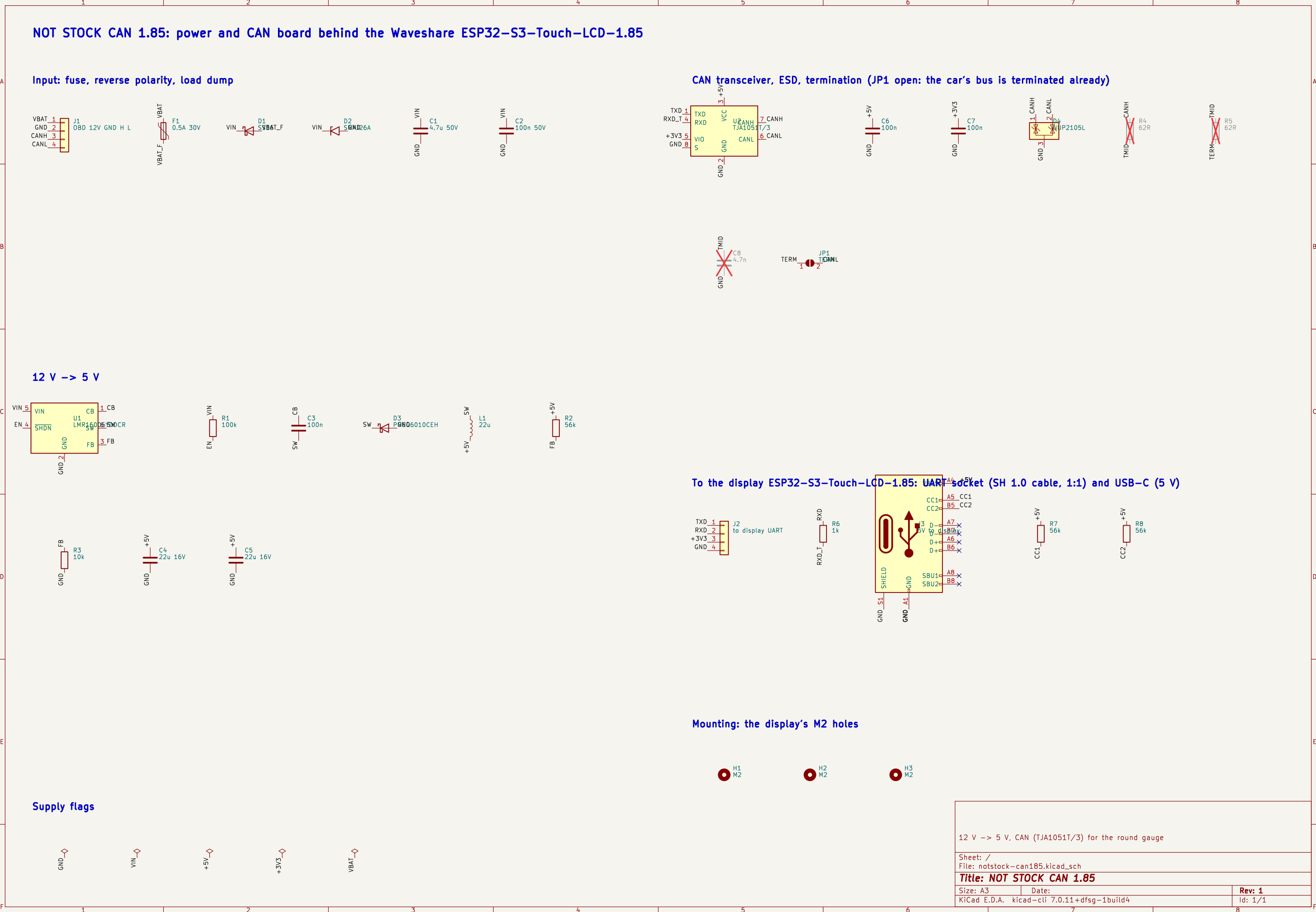
Supply flags

The schematic diagram illustrates the power and CAN bus interface for a Waveshare ESP32-S3-Touch-LCD-1.85 display. The circuit is divided into several functional blocks:

- Input Protection and Power Regulation:** The input section includes a fuse (F1, 0.5A 30V) and a reverse polarity protection diode (D1, 5V BAT_F). The power is then regulated from 12V to 5V using a voltage divider (R1, 100k) and a switch (SW, 0.06010CEH). The output is filtered by a capacitor (C3, 100nF) and a diode (D2, 5V BAT_F).
- Power Filtering and Regulation:** The 5V supply is filtered by a capacitor (C4, 22uF 16V) and a diode (D3, 5V BAT_F). The 3V3 supply is filtered by a capacitor (C5, 22uF 16V) and a diode (D4, 5V BAT_F).
- CAN Transceiver and Termination:** The CAN transceiver (U2, TJA1051T/3) is connected to the CAN bus. It includes termination resistors (R4, 62R, R5, 62R) and a termination jumper (JP1, 1:1). The transceiver is also connected to the CAN bus via a 4.7nF capacitor (C8).
- Power and Ground Connections:** The power and ground connections are labeled: VIN, GND, +5V, +3V3, VBAT, and CANH/CANL.
- Display Connections:** The display's UART socket (J2) is connected to the ESP32-S3-Touch-LCD-1.85. The USB-C port is connected to the display's power and ground pins.
- Mounting:** The display's M2 mounting holes are indicated by H1, H2, and H3.

Supply flags: GND, VIN, +5V, +3V3, VBAT.

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Input: fuse, reverse polarity, load dump

CAN transceiver, ESD, termination (JP1 open: the car's bus is terminated already)

12 V -> 5 V

To the display **ESP32-S3-Touch-LCD-1.85**: UART socket (SH 1.0 cable, 1:1) and USB-C (5 V)

Mounting: the display's M2 holes

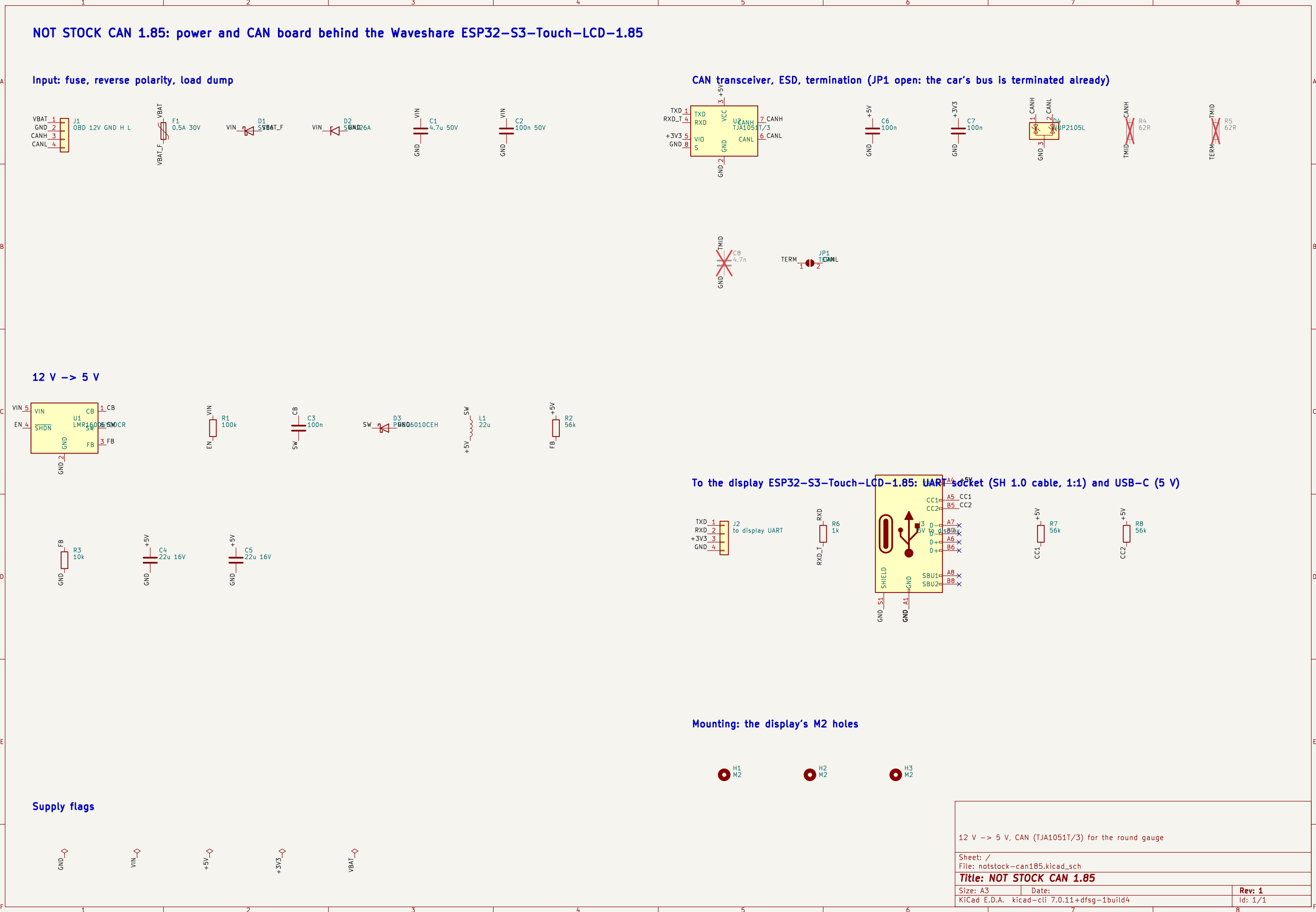
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- Power and Ground Connections:** The power and ground connections are labeled: VIN, GND, +5V, +3V3, VBAT, and CANH/CANL.
- Display Connections:** The display's UART socket (J2) is connected to the ESP32-S3-Touch-LCD-1.85. The USB-C port is connected to the display's power and ground pins.
- Mounting:** The display's M2 mounting holes are indicated by H1, H2, and H3.

Supply flags: GND, VIN, +5V, +3V3, VBAT.

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GND 2
CANH 3
CANL 4

J1 OBD 12V GND H L

VBAT_F VBAT

F1 0.5A 30V

VIN D1 SVD F1 VIN D2 SVD 26A

C1 4.7u 50V

C2 100n 50V

CAN transceiver, ESD, termination (JP1 open: the car's bus is terminated already)

TXD 1 RXD 4 +3V3 5 GND 8

U2 CANH TJA1051T/3 CANL

+5V C6 100n

+3V3 C7 100n

CANH 1 CANL 2

R4 62R

R5 62R

TERM 1 2

12 V -> 5 V

VIN 5 EN 4 CB 1 SW 2 FB 3

U1 LMR16000 DCDC16V

R1 100k

C3 100n

D3 PDS6010CEH

L1 22u

+5V R2 56k

GND 2

R3 10k

+5V C4 22u 16V

+5V C5 22u 16V

To the display ESP32–S3–Touch–LCD–1.85: UART socket (SH 1.0 cable, 1:1) and USB–C (5 V)

TXD 1 RXD 2 +3V3 3 GND 4

J2 to display UART

R6 1k

RXD TXD

CC1 CC2

A5 CC1 B5 CC2

+5V R7 56k

+5V R8 56k

SBU1 SBU2

A8 B8

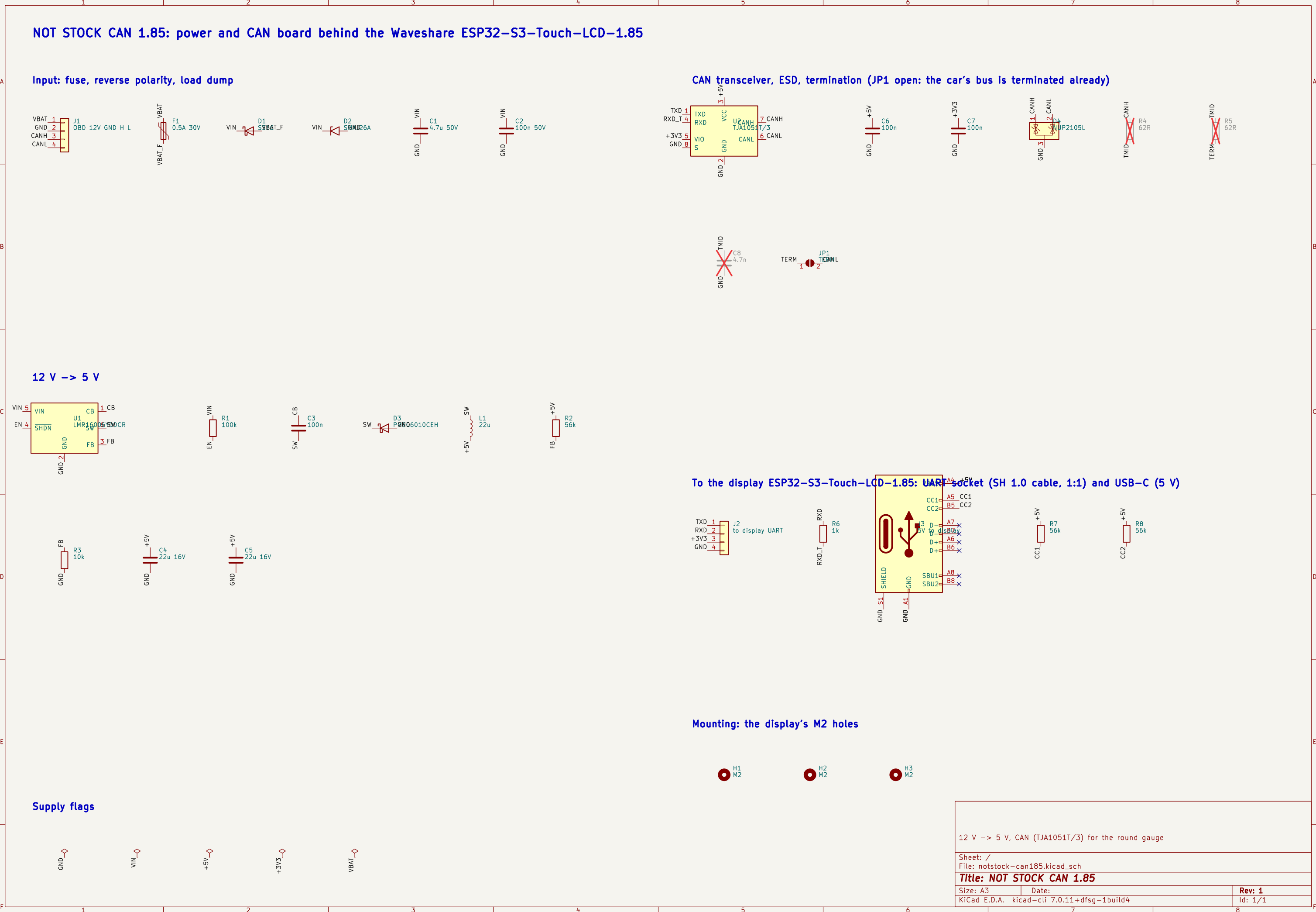
Mounting: the display's M2 holes

H1 M2 H2 M2 H3 M2

Supply flags

GND VIN +5V +3V3 VBAT

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